

Ferroelectric HfO₂ Memory — Instance Example: Flow A and Flow B1

Art Haxholli

A worked example showing what the ontology contains once it is filled with real data. In this case, two flows are shown, and every instance below traces to a specific section or figure of a specific paper.

1 Flow A: laboratory MFM coupon

FlowA_LabMFM is a `Process_Flow`. Source: S1, Section II and Figure 3.

1.1 The process step instances

Seven instances that belong to `Unit_Process` or one of its subclasses.

Instance	is an instance of	Key facts	Source
A1_Substrate_Clean	Wet_Clean	input: Si_substrate	S1
A2_BE_Sputter	Sputter_Deposition	deposits TiN	S1
A3_FE_Deposition	Deposition	deposits HfO2	S1
A4_TE_Capping	Sputter_Deposition	deposits TiN	S1
A5_Crystallisation_Anneal	Crystallisation_Anneal	transforms phase to Orthorhombic_Pca21	S1, Sec. II + Tables I, III, V
A6_Patterned_Metal_Deposition	Deposition	deposits Ti_Pt_contact	S1, Sec. II vs Fig. 3 caption
A7_SC1_Etch	Wet_Etch	etches TiN	S1

Each instance also records which flow it belongs to and which instance comes immediately after it:

$$A1 \rightarrow A2 \rightarrow A3 \rightarrow A4 \rightarrow A5 \rightarrow A6 \rightarrow A7$$

From that chain the ontology derives: **the anneal instance precedes the patterning instance** in this flow. True in flows A, B1, B2, D, F, G; false in Flow C alone.

Why A3_FE_Deposition is typed at the parent class

`Deposition` has subclasses ALD, `Sputter_Deposition`, PLD, CSD. S1 says the ferroelectric can be deposited by any of the four and does not choose.

So `A3_FE_Deposition` is an instance of `Deposition` and of no subclass. If it were made an instance of `ALD`, the ontology would assert something S1 never said, and every later query about ALD would silently include a case that may not be ALD.

Why A6_Patterned_Metal_Deposition is marked as contradicted

S1's body text says step 5 is metal electrode deposition. The Figure 3 caption says step 5 is electrode patterning.

The instance keeps both locators and is marked as contradicted by its own source, S1. Neither is picked as the true version.

1.2 The layer stack instance

Stack_A_MFM is a Layer_Stack. Its layers, bottom to top:

Position	Instance	is an instance of	Role in the stack
1	Si_substrate	Substrate_Material	—
2	SiO2	Dielectric_Layer_Material	—
3	TiN	Electrode_Material	bottom electrode
4	HfO2	Ferroelectric_Material	ferroelectric layer
5	TiN	Electrode_Material	top electrode
6	Ti_Pt_contact	Electrode_Material	contact pad

Ti_Pt_contact is an instance of Electrode_Material because that is where the class tree puts Pt, but in this stack it is a probe contact, so Stack_A_MFM does not link to it through the electrode relations. In PLD and CSD stacks Pt genuinely is a functional electrode.

1.3 Causal instances reachable from this flow

Subject instance	is an instance of	Relation	Object instance	Source
TiN	Electrode_Material	scavenges oxygen from	HfO2	S1, Sec. II A 2
Oxygen_Vacancy	Defect	stabilises phase	Orthorhombic_Pca21, Tetragonal_P42nmc	S1, Sec. I B, II A 2

Orthorhombic_Pca21 and Tetragonal_P42nmc are instances of Crystalline_Phase.

These are reachable from Flow A because A4_TE_Capping puts TiN on HfO2 before A5_Crystallisation_Anneal anneals it. But the ontology does **not** assert that Flow A produced Orthorhombic_Pca21. No source measured a Flow A sample.

1.4 Classes with no instances in Flow A

Class	Instances from Flow A	Why
Measurement	none	S1 reports no P–E loop, endurance or phase fraction for a Flow A sample
Artifact	none	Nothing was measured, so there is no specimen to point at

Class	Instances from Flow A	Why
Metrology_Tool	none	No characterisation, no tool

Also: A5_Crystallisation_Anneal has **no temperature or duration value**. S1's body text gives a range (400–1000 °C, 1–60 s) and its own tables give 1 s to 20 min. See §5.1.

These are empty because the source is silent — not because the answer is zero.

2 Flow B1: gate-last silicon FeFET

FlowB1_GateLast_FeFET is a Process_Flow. Source: S10, Section II A. No technology node recorded.

2.1 The process step instances

Ten instances. Unlike Flow A, every parameter is an exact figure.

Instance	is an instance of	Key facts	Source
B1_1_As_Implant	Implantation	40 keV; $4 \times 10^{15} \text{ cm}^{-2}$	S10
B1_2_Activation_Anneal	Dopant_Activation_Anneal	1050 °C; 5 s; ambient N2	S10
B1_3_dHF_Clean	Wet_Clean	—	S10
B1_4_IL_Ozone_Oxidation	Ozone_Oxidation	300 °C; oxidant O3; uses ALD_reactor	S10
B1_5_HZO_ALD	Thermal_ALD	300 °C; precursors TEMA_Hf, TEMA_Zr; oxidant H2O; uses ALD_reactor	S10
B1_6_TiN_W_Sputter	Sputter_Deposition	deposits TiN, W	S10
B1_7_Crystallisation_Anneal	Crystallisation_Anneal	550 °C; 60 s; ambient N2	S10
B1_8_Gate_Formation	Lithography	—	S10
B1_9_SD_Contact	Contact_Formation	deposits TiN, Al	S10
B1_10_Forming_Gas_Anneal	Forming_Gas_Anneal	450 °C; ambient forming_gas_5H2_95N2	S10

Chain: B1_1 → B1_2 → ... → B1_10.

Two derived facts:

B1_2_Activation_Anneal **precedes** B1_4_IL_Ozone_Oxidation. The gate-last signature. It explains how a flow can contain a 1050 °C step and still produce a film that 1050 °C would destroy — the hot step happens before the ferroelectric exists.

B1_7_Crystallisation_Anneal **precedes** B1_8_Gate_Formation. Anneal before patterning, as in Flow A.

2.2 Supporting instances used by Flow B1

Instances of classes other than `Unit_Process` that the steps above point at.

Instance	is an instance of	Source
TEMA_Hf, TEMA_Zr	Chemical_Precursor	S10
O3, H2O	Oxidant	S10
N2, forming_gas_5H2_95N2	Process_Gas	S10
ALD_reactor	Deposition_Tool	S10
Keysight_B1500A	Electrical_Test_Tool	S10, Sec. II A
Radiant_Precision_LC	Electrical_Test_Tool	S10, Sec. II A

2.3 The layer stack and device instances

`Stack_B1_MFIS` is a `Layer_Stack`.

Position	Instance	is an instance of	Thickness	Role
1	Si_substrate	Channel_Material	—	channel
2	SiO2_IL	Interfacial_Layer_Material	0.7 nm	interfacial layer
3	HZO	Ferroelectric_Material	9 nm	ferroelectric layer
4	TiN_gate	Electrode_Material	10 nm	top electrode
5	W_cap	Electrode_Material	75 nm	gate metal

`Cell_B1_FeFET` is a `FeFET_1T` (a subclass of `Memory_Cell_Architecture`). It uses `Stack_B1_MFIS`, has channel `Si_substrate`, gate length 5 μm , gate width 150 μm .

2.4 Measurement instances

Measurements attach to a **specimen instance**, and they carry the conditions they were taken under.

`Artifact_B1_FeFET_L5_W150` is a `Single_Cell` (a subclass of `Artifact`). This is the physical thing that was measured.

Instance	is an instance of	Points at	Value / conditions	Source
M_B1_Vth	Measurement	method <code>IdVg_Transfer</code> ; property <code>Threshold_Voltage</code> ; measured on <code>Artifact_B1_FeFET_L5_W150</code> ; tool <code>Keysight_B1500A</code>	convention: linear extrapolation	S10, Sec. II A
M_B1_Trap_Density_initial	Measurement	property <code>Trap_Density</code> ; measured on <code>Artifact_B1_FeFET_L5_W150</code>	$1 \times 10^{14} \text{ cm}^{-2}$	S10, Sec. III D
M_B1_Trap_Density_cycled	Measurement	property <code>Trap_Density</code> ; measured on <code>Artifact_B1_FeFET_L5_W150</code>	$3.6 \times 10^{14} \text{ cm}^{-2}$	S10, Sec. III D

`IdVg_Transfer` is an instance of `Measurement_Method`. `Threshold_Voltage` and `Trap_Density` are instances of `Measured_Property`.

Why M_B1_Vth justifies the whole Measurement class

S10 defines threshold voltage by linear extrapolation. S11 defines it by constant current, on a near-identical stack.

If Vth were a plain number stored on HZ0, the S10 and S11 values would look comparable. They are not. Recording the convention on the measurement instance is one field, and it closes the trap permanently.

Likewise the two trap-density figures are **two separate instances**, not one value overwritten — same specimen, different points in its cycling history.

3 Index: every class used, and its instances here

Class	Instances in this example
Process_Flow	FlowA_LabMFM, FlowB1_GateLast_FeFET
Wet_Clean	A1_Substrate_Clean, B1_3_dHF_Clean
Sputter_Deposition	A2_BE_Sputter, A4_TE_Capping, B1_6_TiN_W_Sputter
Deposition (parent only)	A3_FE_Deposition, A6_Patterned_Metal_Deposition
Crystallisation_Anneal	A5_Crystallisation_Anneal, B1_7_Crystallisation_Anneal
Wet_Etch	A7_SC1_Etch
Implantation	B1_1_As_Implant
Dopant_Activation_Anneal	B1_2_Activation_Anneal
Ozone_Oxidation	B1_4_IL_Ozone_Oxidation
Thermal_ALD	B1_5_HZ0_ALD
Lithography	B1_8_Gate_Formation
Contact_Formation	B1_9_SD_Contact
Forming_Gas_Anneal	B1_10_Forming_Gas_Anneal
Layer_Stack	Stack_A_MFM, Stack_B1_MFIS
Electrode_Material	TiN, Ti_Pt_contact, TiN_gate, W_cap
Ferroelectric_Material	HfO2, HZ0
Interfacial_Layer_Material	SiO2_IL
Dielectric_Layer_Material	SiO2
Channel_Material	Si_substrate
Substrate_Material	Si_substrate
Chemical_Precursor	TEMA_Hf, TEMA_Zr
Oxidant	O3, H2O
Process_Gas	N2, forming_gas_5H2_95N2
Crystalline_Phase	Orthorhombic_Pca21, Tetragonal_P42nmc
Defect	Oxygen_Vacancy
Deposition_Tool	ALD_reactor
Electrical_Test_Tool	Keysight_B1500A, Radiant_Precision_LC

Class	Instances in this example
Memory_Cell_Architecture → FeFET_1T	Cell_B1_FeFET
Artifact → Single_Cell	Artifact_B1_FeFET_L5_W150
Measurement_Method	IdVg_Transfer
Measured_Property	Threshold_Voltage, Trap_Density
Measurement	M_B1_Vth, M_B1_Trap_Density_initial, M_B1_Trap_Density_cycled

Classes deliberately left empty by Flow A: Measurement, Artifact, Metrology_Tool.

⇒ **UNCERTAIN** Si_substrate appears under two classes because Flow A uses it as a substrate and Flow B1 uses it as a channel. Either it is one instance with two types, or the two flows need distinct instances.